A ONE-POINT DM NOETHERIAN REDUCED STACK ADMITS A SURJECTIVE ETALE MORPHISM FROM THE SPECTRUM OF A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The description of a presentation of a DM stack by an etale groupoid with etale diagonal. Used inside the counted proof (source lines 4413-4430). Supplies the smooth presentation and the etale comparison morphisms used throughout.
- The criterion for a morphism of algebraic spaces to be an open immersion through etaleness and universal injectivity. Invoked inside the counted proof (source lines 4440-4444) as a cross-chapter pointer. Turns the etale diagonal of the presentation into an open immersion.
- The characterization of unramifiedness through the diagonal being an open immersion. Invoked inside the counted proof (source lines 4445-4448) as a cross-chapter pointer. Concludes unramifiedness of the comparison morphism.

Source reference. Original reference algebraic-lemma-criterion-map-representable-spaces-fibred-in-groupoids: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 431–431. Navigation pointer only: the counted text refers to the Stacks item algebraic-lemma-criterion-map-representable-spaces-fibred-in-groupoids, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference algebraic-remark-flat-fp-presentation: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4422–4422. Navigation pointer only: the counted text refers to the Stacks item algebraic-remark-flat-fp-presentation, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference bootstrap-lemma-divide-subgroupoid: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4491–4491. Navigation pointer only: the counted text refers to the Stacks item bootstrap-lemma-divide-subgroupoid, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference bootstrap-theorem-final-bootstrap: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4487–4487. Navigation pointer only: the counted text refers to the Stacks item bootstrap-theorem-final-bootstrap, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference descent-lemma-descending-property-flat: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4511–4511. Navigation pointer only: the counted text refers to the Stacks item descent-lemma-descending-property-flat, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference descent-lemma-descending-property-locally-finite-presentation: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4512–4512. Navigation pointer only: the counted text refers to the Stacks item descent-lemma-descending-property-locally-finite-presentation, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference descent-lemma-descending-property-open-immersion: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4525–4525. Navigation pointer only: the counted text refers to the Stacks item descent-lemma-descending-property-open-immersion, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference descent-lemma-descending-property-quasi-finite: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4513–4513. Navigation pointer only: the counted text refers to the Stacks item descent-lemma-descending-property-quasi-finite, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference etale-lemma-universally-injective-unramified: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4462–4462. Navigation pointer only: the counted text refers to the Stacks item etale-lemma-universally-injective-unramified, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference groupoids-equation-invariants: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4472–4472. Navigation pointer only: the counted text refers to the Stacks item groupoids-equation-invariants, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference groupoids-proposition-finite-flat-equivalence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4465–4465. Navigation pointer only: the counted text refers to the Stacks item groupoids-proposition-finite-flat-equivalence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference more-groupoids-lemma-sheaf-differentials: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4532–4532. Navigation pointer only: the counted text refers to the Stacks item more-groupoids-lemma-sheaf-differentials, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference morphisms-lemma-flat-unramified-etale: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4539–4539. Navigation pointer only: the counted text refers to the Stacks item morphisms-lemma-flat-unramified-etale, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference morphisms-lemma-unramified-omega-zero: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4536–4536. Navigation pointer only: the counted text refers to the Stacks item morphisms-lemma-unramified-omega-zero, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-groupoids-lemma-quotient-stack-restrict-equivalence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4503–4503. Navigation pointer only: the counted text refers to the Stacks item spaces-groupoids-lemma-quotient-stack-restrict-equivalence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-more-groupoids-lemma-groupoid-on-field-dimension-equal-stabilizer: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4333–4333. Navigation pointer only: the counted text refers to the Stacks item spaces-more-groupoids-lemma-groupoid-on-field-dimension-equal-stabilizer, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-over-fields-lemma-locally-finite-type-dim-zero: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4438–4438. Navigation pointer only: the counted text refers to the Stacks item spaces-over-fields-lemma-locally-finite-type-dim-zero, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-check-property-covering: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 479–479. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-check-property-covering, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-flat-cover-by-field: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4482–4482. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-flat-cover-by-field, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-unique-point-better: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4414–4414. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-unique-point-better, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference varieties-lemma-algebraic-scheme-dim-0: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4440–4440. Navigation pointer only: the counted text refers to the Stacks item varieties-lemma-algebraic-scheme-dim-0, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{Z}$ be a DM, locally Noetherian, reduced algebraic stack with $|\mathcal{Z}|$ a singleton. Then there exists a field k and a surjective étale morphism $\mathrm{Spec}(k) \to \mathcal{Z}$.*

Proof. By Properties of Stacks, Lemma stacks-properties-lemma-unique-point-better there exists a field k and a surjective, flat, locally finitely presented morphism

$\mathrm{Spec}(k) \to \mathcal{Z}$. Set $U = \mathrm{Spec}(k)$ and $R = U \times_{\mathcal{Z}} U$ so we obtain a groupoid in algebraic spaces (U, R, s, t, c), see Algebraic Stacks, Lemma algebraic-lemma-criterion-map-representable-spaces-fibred-in-groupoids. Note that by Algebraic Stacks, Remark algebraic-remark-flat-fp-presentation we have an equivalence

$$f_{can} : [U/R] \longrightarrow \mathcal{Z}$$

The projections $s, t : R \to U$ are locally of finite presentation. As $\mathcal{Z}$ is DM we see that the stabilizer group algebraic space

$$G = U \times_{U \times U} R = U \times_{U \times U} (U \times_{\mathcal{Z}} U) = U \times_{\mathcal{Z} \times \mathcal{Z}, \Delta_{\mathcal{Z}}} \mathcal{Z}$$

is unramified over U. In particular $\dim(G) = 0$ and by More on Groupoids in Spaces, Lemma spaces-more-groupoids-lemma-groupoid-on-field-dimension-equal-stabilizer we have $\dim(R) = 0$. This implies that R is a scheme, see Spaces over Fields, Lemma spaces-over-fields-lemma-locally-finite-type-dim-zero. By Varieties, Lemma varieties-lemma-algebraic-scheme-dim-0 we see that R (and also G) is the disjoint union of spectra of Artinian local rings finite over k via either s or t. Let $P = \mathrm{Spec}(A) \subset R$ be the open and closed subscheme whose underlying point is the identity e of the groupoid scheme (U, R, s, t, c). As $s \circ e = t \circ e = \mathrm{id}_{\mathrm{Spec}(k)}$ we see that A is an Artinian local ring whose residue field is identified with k via either $s^\sharp : k \to A$ or $t^\sharp : k \to A$. Note that $s, t : \mathrm{Spec}(A) \to \mathrm{Spec}(k)$ are finite (by the lemma referenced above). Since $G \to \mathrm{Spec}(k)$ is unramified we see that

$$G \cap P = P \times_{U \times U} U = \mathrm{Spec}(A \otimes_{k \otimes k} k)$$

is unramified over k. On the other hand $A \otimes_{k \otimes k} k$ is local as a quotient of A and surjects onto k. We conclude that $A \otimes_{k \otimes k} k = k$. It follows that $P \to U \times U$ is universally injective (as P has only one point with residue field k), unramified (by the computation of the fibre over the unique image point above), and of finite type (because s, t are) hence a monomorphism (see Étale Morphisms, Lemma etale-lemma-universally-injective-unramified). Thus $s|_P, t|_P : P \to U$ define a finite flat equivalence relation. Thus we may apply Groupoids, Proposition groupoids-proposition-finite-flat-equivalence to conclude that U/P exists and is a scheme $\overline{U}$. Moreover, $U \to \overline{U}$ is finite locally free and $P = U \times_{\overline{U}} U$. In fact $\overline{U} = \mathrm{Spec}(k_0)$ where $k_0 \subset k$ is the ring of R-invariant functions. As k is a field it follows from the definition Groupoids, Equation (groupoids-equation-invariants) that k_0 is a field.

We claim that

$$\mathrm{Spec}(k_0) = \overline{U} = U/P \to [U/R] = \mathcal{Z} \tag{0.1.1}$$

is the desired surjective étale morphism. It follows from Properties of Stacks, Lemma stacks-properties-lemma-flat-cover-by-field that this morphism is surjective. Thus it suffices to show that (0.1.1) is étale[1]. Instead of proving the étaleness directly we first apply Bootstrap, Lemma bootstrap-lemma-divide-subgroupoid to see that there exists a groupoid scheme $(\overline{U}, \overline{R}, \overline{s}, \overline{t}, \overline{c})$ such that (U, R, s, t, c) is the restriction of $(\overline{U}, \overline{R}, \overline{s}, \overline{t}, \overline{c})$ via the quotient morphism $U \to \overline{U}$. (We verified all the hypothesis of the lemma above except for the assertion that $j : R \to U \times U$ is separated and locally quasi-finite which follows from the fact that R is a separated scheme locally quasi-finite over k.) Since $U \to \overline{U}$ is finite locally free we see that $[U/R] \to [\overline{U}/\overline{R}]$ is an equivalence, see Groupoids in Spaces, Lemma spaces-groupoids-lemma-quotient-stack-restrict-equivalence.

Note that s, t are the base changes of the morphisms $\overline{s}, \overline{t}$ by $U \to \overline{U}$. As $\{U \to \overline{U}\}$ is an fppf covering we conclude $\overline{s}, \overline{t}$ are flat, locally of finite presentation, and

[1] We urge the reader to find his/her own proof of this fact. In fact the argument has a lot in common with the final argument of the proof of Bootstrap, Theorem bootstrap-theorem-final-bootstrap hence probably should be isolated into its own lemma somewhere.

locally quasi-finite, see Descent, Lemmas descent-lemma-descending-property-flat, descent-lemma-descending-property-locally-finite-presentation, and descent-lemma-descending-property-quasi-finite. Consider the commutative diagram

$$\begin{array}{ccccc} U \times_{\overline{U}} U & = & P & \longrightarrow & R \\ & \searrow & \downarrow & & \downarrow \\ & & \overline{U} & \xrightarrow{\overline{e}} & \overline{R} \end{array}$$

It is a general fact about restrictions that the outer four corners form a cartesian diagram. By the equality we see the inner square is cartesian. Since P is open in R we conclude that $\overline{e}$ is an open immersion by Descent, Lemma descent-lemma-descending-property-open-immersion.

But of course, if $\overline{e}$ is an open immersion and $\overline{s}, \overline{t}$ are flat and locally of finite presentation then the morphisms $\overline{t}, \overline{s}$ are étale. For example you can see this by applying More on Groupoids, Lemma more-groupoids-lemma-sheaf-differentials which shows that $\Omega_{\overline{R}/\overline{U}} = 0$ implies that $\overline{s}, \overline{t} : \overline{R} \to \overline{U}$ is unramified (see Morphisms, Lemma morphisms-lemma-unramified-omega-zero), which in turn implies that $\overline{s}, \overline{t}$ are étale (see Morphisms, Lemma morphisms-lemma-flat-unramified-etale). Hence $\mathcal{Z} = [\overline{U}/\overline{R}]$ is an étale presentation of the algebraic stack $\mathcal{Z}$ and we conclude that $\overline{U} \to \mathcal{Z}$ is étale by Properties of Stacks, Lemma stacks-properties-lemma-check-property-covering. □